
Concept Questions: Chapter 8

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ME-3640 – Heat Transfer

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Abstract

Fundamental principals and concepts are far more important to engineering expertise than understanding of example systems. These concept questions (shamelessly lifted from the textbook) will help you test your knowledge after reading the chapter summary and building a concept map. Use the relative ease and difficulty you experience while answering these questions to prioritize your efforts fo the remainder of this chapter.

1 General Flow Analysis

- 8-1C** Which fluid at room temperature requires a larger pump to move at a specified velocity in a given tube: water or engine oil? Why?
- 8-2C** What do the average velocity V_{avg} and the mean temperature T_m represent in flow through circular tubes of constant diameter?
- 8-3C** What is the generally accepted value of the Reynolds number above which the flow in smooth pipes is turbulent?
- 8-4C** What is hydraulic diameter? How is it defined? What is it equal to for a circular tube of diameter D ?
- 8-5C** What fluid property is responsible for the development of the velocity boundary layer? For what kinds of fluids will there be no velocity boundary layer in a pipe?
- 8-6C** In the fully developed region of flow in a circular tube, will the velocity profile change in the flow direction? How about the temperature profile?
- 8-7C** Consider laminar flow in a circular tube. Will the friction factor be higher near the inlet of the tube or near the exit? Why? What would your response be if the flow were turbulent?
- 8-8C** How does the friction factor f vary along the flow direction in the fully developed region in (a) laminar flow and (b) turbulent flow?
- 8-9C** How does surface roughness affect the pressure drop in a tube if the flow is turbulent? What would your response be if the flow were laminar?
- 8-10C** How is the hydrodynamic entry length defined for flow in a tube? Is the entry length longer in laminar or turbulent flow?
- 8-11C** Consider the flow of oil in a tube. How will the hydrodynamic and thermal entry lengths compare if the flow is laminar? How would they compare if the flow were turbulent?
- 8-12C** Consider the flow of mercury (a liquid metal) in a tube. How will the hydrodynamic and thermal entry lengths compare if the flow is laminar? How would they compare if the flow were turbulent?
- 8-13C** How is the thermal entry length defined for flow in a tube? In what region is the flow in a tube fully developed?
- 8-14C** Consider laminar forced convection in a circular tube. Will the heat flux be higher near the inlet of the tube or near the exit? Why?
- 8-15C** Consider turbulent forced convection in a circular tube. Will the heat flux be higher near the inlet of the tube or near the exit? Why?

- 8-16C** What does the logarithmic mean temperature difference represent for flow in a tube whose surface temperature is constant? Why do we use the logarithmic mean temperature instead of the arithmetic mean temperature?
- 8-17C** Consider fluid flow in a tube whose surface temperature remains constant. What is the appropriate temperature difference for use in Newton's law of cooling with an average heat transfer coefficient?
- 8-18C** What is the physical significance of the number of transfer units $NTU = hA_s/\dot{m}C_p$? What do small and large NTU values tell about a heat transfer system?

2 Laminar and Turbulent Flow in Tubes

- 8-31C** Someone claims that the volume flow rate in a circular pipe with laminar flow can be determined by measuring the velocity at the centerline in the fully developed region, multiplying it by the cross-sectional area, and dividing the result by 2. Do you agree? Explain.
- 8-32C** Someone claims that the average velocity in a circular pipe in fully developed laminar flow can be determined by simply measuring the velocity at $R/2$ (midway between the wall surface and the centerline). Do you agree? Explain.
- 8-33C** How is the friction factor for flow in a tube related to the pressure drop? How is the pressure drop related to the pumping power requirement for a given mass flow rate?
- 8-34C** Someone claims that in fully developed turbulent flow in a tube, the shear stress is a maximum at the tube surface. Do you agree with this claim? Explain.
- 8-35C** Consider fully developed flow in a circular pipe with negligible entrance effects. If the length of the pipe is doubled, the pressure drop will (a) double, (b) more than double, (c) less than double, (d) reduce by half, or (e) remain constant.
- 8-36C** Consider fully developed laminar flow in a circular pipe. If the diameter of the pipe is reduced by half while the flow rate and the pipe length are held constant, the pressure drop will (a) double, (b) triple, (c) quadruple, (d) increase by a factor of 8, or (e) increase by a factor of 16.
- 8-37C** Consider fully developed laminar flow in a circular pipe. If the viscosity of the fluid is reduced by half by heating while the flow rate is held constant, how will the pressure drop change?

A Rubrics

Submissions of this assignment will be graded upon the criteria found in this section. Review these carefully, and complete your work so that application of this rubric produces your desired score.

 Criterion 1 Problems	
<i>Homework is completed with appropriate effort, neatness, etc.</i>	
Points	Description
20	Honest effort to participate and learn: every problem shows an answer or questions which need resolution.
10	Rushed or haphazard work: some problems show no effort at all.
0	Work is absent, wholly incorrect, or plagiarized.